Igor Steinmacher, PhD

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Highlights

- Currently supervising 3 Ph.D. students
- Successfully supervised 20+ undergraduate research students, 9 M.Sc. and 11 Ph.D. students (as supervisor and co-supervisor)
- Supervisor of Bianca Trinkenreich's PhD dissertation, winner of the 2024 ACM SIGSOFT Outstanding Dissertation Award
- Best Paper / Distinguished Paper Awards at CHASE@ICSE 2026, ESEM 2024, ICSE 2023, CONVERSATIONS 2023, ICSE-SEIS 2022, ICSE 2021, ICSME 2020, OpenSym 2020, ICSE-SEET 2019, SBES 2018
- Higher education instructor since 2005
- Secured more than USD 2.5M as lead PI of NSF grants
- Lead guest editor of the Thematic Series on Free and Open Source Software Development, SpringerOpen Journal of Internet Services and Applications (JISA)
- Guest editor of the IEEE Software special issue “20 Years of Open Source—Impact on Software Engineering Practice”
- Citation index h44 (Google Scholar)

Education

- **2015** — Ph.D. in Computer Science, University of São Paulo (USP), Brazil
- **2005** — M.Sc. in Computer Science, Federal University of Rio Grande do Sul (UFRGS), Brazil
- **2003** — B.S. in Computer Science, State University of Maringá (UEM), Brazil

Professional Experience

- **2025–current** — Associate Professor & Associate Director for Graduate Programs and Academic Affairs, Northern Arizona University (NAU), School of Informatics, Computing and Cyber Systems, USA
- **2018–2025** — Assistant Professor, Northern Arizona University (NAU), School of Informatics, Computing and Cyber Systems, USA (on leave of absence from Jan 2021 to Jul 2022)
- **2017–2018** — Post-doctoral Scholar, Northern Arizona University (NAU), USA
- **2009–2021** — Faculty Member, Federal University of Technology – Paraná (UTFPR), Department of Computing, Brazil (on leave of absence from Jan 2017 to Dec 2020)
- **2013–2014** — Visiting Scholar, University of California, Irvine (UCI), Department of Informatics, USA
- **2006–2009** — Project Manager, HSBC Global Technology, Brazil
- **2006–2009** — Lecturer, Faculdades Santa Cruz, Brazil
- **2005–2006** — Software Developer, CELEPAR – Data Processing Company of Paraná, Brazil

Grants

- **CommUnityBuddy** — a conversational agent supporting open source contributors. Alfred P. Sloan Foundation, 2026–current, lead PI, USD 100K.
- **NSF POSE: expanding the data.table ecosystem** — for efficient big data manipulation in R. 2023–2027, lead PI, USD 731K, with Marco Gerosa and Toby Hocking.
- **NSF HSI: a learning environment for an open-source contribution model** — OSS-Doorway. 2023–2027, lead PI, USD 498K, with Marco Gerosa.
- **NSF RETTL: learning software engineering by contributing to real projects with chatbot assistance**. 2023–2027, co-PI, USD 562K, with Marco Gerosa (PI).
- **NSF IUSE: scaffolding computational thinking in introductory CS through a conversational agent** — SIGIL. 2023–2026, lead PI, USD 405K, with Marco Gerosa.
- **NSF CHS: Gender-Inclusive Open Source through Gender-Inclusive Tools**. 2019–2023, lead PI, USD 528K.
- **Fostering open source communities by supporting newcomer developers**. FAPESP / MCTI CGI.br (Brazil), 2017, lead PI, ~BRL 230K, with Igor Wiese and Ivanilton Polato.

Awards and Honors

- **2026** — ACM SIGSOFT Distinguished Paper — CHASE@ICSE 2026: “Governance in Practice: How Open Source Projects Define and Document Roles”
- **2024** — Best Paper Award — ESEM 2024: “ChatGPT application in Systematic Literature Reviews in Software Engineering: an evaluation of its accuracy to support the selection activity”
- **2024** — ACM SIGSOFT Outstanding Dissertation Award — awarded to Bianca Trinkenreich, my PhD advisee
- **2023** — ACM SIGSOFT Distinguished Paper — ICSE 2023: “Do I Belong? Modeling Sense of Virtual Community Among Linux Kernel Contributors”
- **2023** — Best Paper Award — CONVERSATIONS 2023: “Anticipating User Needs: Insights from Design Fiction on Conversational Agents for Computational Thinking”
- **2023** — Distinguished Reviewer Award — ICSE 2023
- **2022** — Best Paper Award — ICSE-SEIS 2022: “An Empirical Investigation on the Challenges Faced by Women in the Software Industry: A Case Study”
- **2021** — ACM SIGSOFT Distinguished Paper — ICSE 2021: “What Makes a Great Maintainer of Open Source Projects?”
- **2021–2022** — Brazilian Research Productivity Fellowship (PQ)
- **2020** — IEEE TCSE Distinguished Paper — ICSME 2020: “Effects of Adopting Code Review Bots on Pull Requests to OSS Projects”
- **2020** — Best Paper Award — OpenSym 2020: “Recommending Tasks to Newcomers in OSS Projects: How Do Mentors Handle It?”
- **2020** — Honorable Mention — CSCW 2020: “Hidden Figures: Roles and Pathways of Successful OSS Contributors”
- **2019** — ACM SIGSOFT Distinguished Paper — ICSE-SEET 2019: “Training software engineers using open-source software: the students' perspective”
- **2018** — Best Paper Award — SBES 2018: “An empirical study on task documentation in software crowdsourcing: the case of the topcoder platform”

Publications

57 journal articles and 147 conference and workshop papers, from DBLP. ★ marks award-winning papers.

Journal Articles

- Daniel Coutinho, Breno Braga, Theo Canuto, Juliana Alves Pereira, Wesley K. G. Assunção, **Igor Steinmacher**, Marco Aurélio Gerosa, Alessandro Garcia. Leveraging large language models for sentiment analysis in GitHub pull request discussions. *Empir. Softw. Eng.* 31(5): 140 (2026).
- João Vitor Souza Rocha, Igor Scaliante Wiese, Ivanilton Polato, Marco Aurélio Graciotto Silva, Reginaldo Ré, **Igor Steinmacher**, Walter Takashi Nakamura. Investigating the potential of using worked examples to help resolve issues in a GitHub project. *J. Syst. Softw.* 236: 112810 (2026).
- Genildo Gomes, Fernando Nogueira, João Gustavo Kienen, Bernardo Scarambone, Desiree Scarambone, Leandro Galvão, Tanara Lauschner, **Igor Steinmacher**, Tayana Conte, Bruno Gadelha. FeelIt: A Tool for Expressing Real-Time Emotions and Its Influence on the Audience Experience. *J. Braz. Comput. Soc.* 32(1): 1812-1826 (2026).
- Zixuan Feng, Reed Milewicz, Emerson Murphy-Hill, Tyler Menezes, Alexander Serebrenik, **Igor Steinmacher**, Anita Sarma. Charting Uncertain Waters: A Socio-Technical Roadmap for Sustaining Open Source Communities in the Age of GenAI. *ACM Trans. Softw. Eng. Methodol.* 35(8): 214:1-214:21 (2026).
- Doris Amoakohene, Anirban Chetia, **Igor Steinmacher**, Toby Hocking. Asymptotic Benchmarking Using the atime package. *R J.* 18(1): 169-187 (2026).
- Zixuan Feng, Katie Kimura, Bianca Trinkenreich, **Igor Steinmacher**, Marco Aurélio Gerosa, Anita Sarma. Addressing OSS Community Managers' Challenges in Contributor Retention. *ACM Trans. Softw. Eng. Methodol.* 35(7): 187:1-187:38 (2026).
- Ítalo Santos, Kátia Romero Felizardo, **Igor Steinmacher**, Marco Aurélio Gerosa. Software solutions for newcomers' onboarding in software projects: A systematic literature review. *Inf. Softw. Technol.* 177: 107568 (2025).
- Márcia Lima, **Igor Steinmacher**, Denae Ford, Grace Vorreuter, Ludimila Gonçalves, Tayana Conte, Bruno Gadelha. How are discussions linked? A link analysis study on GitHub Discussions. *J. Syst. Softw.* 219: 112196 (2025).
- Bianca Trinkenreich, Marco Aurélio Gerosa, Anita Sarma, **Igor Steinmacher**. Guidelines for Cultivating a Sense of Belonging to Reduce Developer Burnout. *IEEE Softw.* 42(2): 84-91 (2025).
- Carolline Pena, Bruno Cartaxo, **Igor Steinmacher**, Deepika Badampudi, Deyvson da Silva, Williby da Silva Ferreira, Adauto Almeida, Fernando Kamei, Sérgio Soares. Comparing the Efficacy of Rapid Review With a Systematic Review in the Software Engineering Field. *J. Softw. Evol. Process.* 37(1) (2025).
- Gabriel Aracena, Kyle Luster, Fabio Santos, **Igor Steinmacher**, Marco Aurélio Gerosa. Applying large language models to issue classification: Revisiting with extended data and new models. *Sci. Comput. Program.* 246: 103333 (2025).
- Camila A. Paiva, Raquel Maximino, Frederico Paiva, Rafael Accetta Vieira, Nicole Espanha, João Felipe Pimentel, Igor Wiese, Marco Aurélio Gerosa, **Igor Steinmacher**, Leonardo Murta, Vanessa Braganholo. Analyzing the adoption of database management systems throughout the history of open source projects. *Empir. Softw. Eng.* 30(3): 71 (2025).
- Zixuan Feng, Katie Kimura, Bianca Trinkenreich, Anita Sarma, **Igor Steinmacher**. Guiding the way: A systematic literature review on mentoring practices in open source software projects. *Inf. Softw. Technol.* 171: 107470 (2024).
- Martin P. Robillard, Deeksha M. Arya, Neil A. Ernst, Jin L. C. Guo, Maxime Lamothe, Mathieu Nassif, Nicole Novielli, Alexander Serebrenik, **Igor Steinmacher**, Klaas-Jan Stol. Communicating Study Design Trade-offs in Software Engineering. *ACM Trans. Softw. Eng. Methodol.* 33(5): 112:1-112:10 (2024).
- Marco Aurélio Gerosa, Bianca Trinkenreich, **Igor Steinmacher**, Anita Sarma. Can AI serve as a substitute for human subjects in software engineering research? *Autom. Softw. Eng.* 31(1): 13 (2024).

- Simone de França Tonhão, Thelma Elita Colanzi, **Igor Steinmacher**. A portal to catalog worked examples extracted from open source software projects to support the teaching of Software Engineering. *J. Softw. Eng. Res. Dev.* 12(1) (2024).
- Fabio Santos, Joseph Vargovich, Bianca Trinkenreich, Ítalo Santos, Jacob Penney, Ricardo Britto, João Felipe Pimentel, Igor Wiese, **Igor Steinmacher**, Anita Sarma, Marco Aurélio Gerosa. Tag that issue: applying API-domain labels in issue tracking systems. *Empir. Softw. Eng.* 28(5): 116 (2023).
- Patrícia G. F. Matsubara, **Igor Steinmacher**, Bruno Gadelha, Tayana Conte. Much more than a prediction: Expert-based software effort estimation as a behavioral act. *Empir. Softw. Eng.* 28(4): 98 (2023).
- Márcia Lima, **Igor Steinmacher**, Denae Ford, Evangeline Liu, Grace Vorreuter, Tayana Conte, Bruno Gadelha. Looking for related posts on GitHub discussions. *PeerJ Comput. Sci.* 9: e1567 (2023).
- Mairieli Wessel, Andy Zaidman, Marco Aurélio Gerosa, **Igor Steinmacher**. Guidelines for Developing Bots for GitHub. *IEEE Softw.* 40(3): 72-79 (2023).
- **Igor Steinmacher**, Paul M. Clarke, Eray Tuzun, Ricardo Britto. Editorial: Machine learning, software process, and global software engineering. *J. Softw. Evol. Process.* 35(6) (2023).
- **Igor Steinmacher**, Paul M. Clarke, Eray Tuzun, Ricardo Britto. Editorial: Best papers of the 14th International Conference on Software and System Processes (ICSSP 2020) and 15th International Conference on Global Software Engineering (ICGSE 2020). *J. Softw. Evol. Process.* 35(5) (2023).
- Bianca Trinkenreich, Igor Wiese, Anita Sarma, Marco Aurélio Gerosa, **Igor Steinmacher**. Women's Participation in Open Source Software: A Survey of the Literature. *ACM Trans. Softw. Eng. Methodol.* 31(4): 81:1-81:37 (2022).
- Bianca Trinkenreich, Marco Aurélio Gerosa, **Igor Steinmacher**. Women in Open Source: We Need to Talk About It. *Computer* 55(12): 145-149 (2022).
- Fabio Calefato, Marco Aurélio Gerosa, Giuseppe Iaffaldano, Filippo Lanubile, **Igor Steinmacher**. Will you come back to contribute? Investigating the inactivity of OSS core developers in GitHub. *Empir. Softw. Eng.* 27(3): 76 (2022).
- **Igor Steinmacher**, Rafael Prikladnicki, Tayana Conte. Summary of the 12th Brazilian Conference on Software and the 1st Latin American School on Software Engineering. *ACM SIGSOFT Softw. Eng. Notes* 47(3): 28-31 (2022).
- Patrícia Gomes Fernandes Matsubara, Bruno F. Gadelha, **Igor Steinmacher**, Tayana Uchôa Conte. SEXTAMT: A systematic map to navigate the wide seas of factors affecting expert judgment software estimates. *J. Syst. Softw.* 185: 111148 (2022).
- Mairieli Santos Wessel, Alexander Serebrenik, Igor Wiese, **Igor Steinmacher**, Marco Aurélio Gerosa. Quality gatekeepers: investigating the effects of code review bots on pull request activities. *Empir. Softw. Eng.* 27(5): 108 (2022).
- Bianca Trinkenreich, Mariam Guizani, Igor Wiese, Tayana Conte, Marco Aurélio Gerosa, Anita Sarma, **Igor Steinmacher**. Pots of Gold at the End of the Rainbow: What is Success for Open Source Contributors? *IEEE Trans. Software Eng.* 48(10): 3940-3953 (2022).
- Hema Susmita Padala, Christopher J. Mendez, Felipe Fronchetti, **Igor Steinmacher**, Zoe Steine-Hanson, Claudia Hilderbrand, Amber Horvath, Charles Hill, Logan Simpson, Margaret Burnett, Marco Aurélio Gerosa, Anita Sarma. How Gender-Biased Tools Shape Newcomer Experiences in OSS Projects. *IEEE Trans. Software Eng.* 48(2): 241-259 (2022).
- Raula Gaikovina Kula, Christoph Treude, Hideaki Hata, Sebastian Baltes, **Igor Steinmacher**, Marco Aurélio Gerosa, Winifred Kula Amini. Challenges for Inclusion in Software Engineering: The Case of the Emerging Papua New Guinean Society. *IEEE Softw.* 39(3): 67-76 (2022).
- Mariam Guizani, Amreeta Chatterjee, Bianca Trinkenreich, Mary Evelyn May, Geraldine J. Noa-Guevara, Liam James Russell, Griselda G. Cuevas Zambrano, Daniel Izquierdo-Cortazar, **Igor Steinmacher**, Marco Aurélio Gerosa, Anita Sarma. The Long Road Ahead: Ongoing Challenges in Contributing to Large OSS Organizations and What to Do. *Proc. ACM Hum. Comput. Interact.* 5(CSCW2): 407:1-407:30 (2021).

- Flavia Bernardini, André Grützmann, **Igor Steinmacher**, Awdren L. Fontão, Davi Viana, Flávio E. A. Horita. Editorial Vol. 14 No. 3. *Braz. J. Inf. Syst.* 14(3): 1-4 (2021).
- Mairieli Santos Wessel, Igor Wiese, **Igor Steinmacher**, Marco Aurélio Gerosa. Don't Disturb Me: Challenges of Interacting with Software Bots on Open Source Software Projects. *Proc. ACM Hum. Comput. Interact.* 5(CSCW2): 301:1-301:21 (2021).
- **Igor Steinmacher**, Sogol Balali, Bianca Trinkenreich, Mariam Guizani, Daniel Izquierdo-Cortazar, Griselda G. Cuevas Zambrano, Marco Aurélio Gerosa, Anita Sarma. Being a Mentor in open source projects. *J. Internet Serv. Appl.* 12(1): 7 (2021).
- Paolo Tell, David Raffo, Liguo Huang, **Igor Steinmacher**, Ricardo Britto, Eray Tüzün, Paul M. Clarke. Summary of the 1st ICSSP-ICGSE Joint Event. *ACM SIGSOFT Softw. Eng. Notes* 45(4): 31-34 (2020).
- Bianca Trinkenreich, Mariam Guizani, Igor Wiese, Anita Sarma, **Igor Steinmacher**. Hidden Figures: Roles and Pathways of Successful OSS Contributors. *Proc. ACM Hum. Comput. Interact.* 4(CSCW2): 180:1-180:22 (2020). ★ Honorable Mention
- Jefferson De Oliveira Silva, Igor Wiese, Daniel M. Germán, Christoph Treude, Marco Aurélio Gerosa, **Igor Steinmacher**. Google summer of code: Student motivations and contributions. *J. Syst. Softw.* 162 (2020).
- Edson Oliveira, Eduardo Fernandes, **Igor Steinmacher**, Marco Cristo, Tayana Conte, Alessandro Garcia. Code and commit metrics of developer productivity: a study on team leaders perceptions. *Empir. Softw. Eng.* 25(4): 2519-2549 (2020).
- Gregorio Robles, **Igor Steinmacher**, Paul Adams, Christoph Treude. Twenty Years of Open Source Software: From Skepticism to Mainstream. *IEEE Softw.* 36(6): 12-15 (2019).
- Dan Sholler, **Igor Steinmacher**, Denae Ford, Mara Averick, Mike Hoye, Greg Wilson. Ten simple rules for helping newcomers become contributors to open projects. *PLoS Comput. Biol.* 15(9) (2019).
- Williamson Silva, **Igor Steinmacher**, Tayana Conte. Students' and Instructors' Perceptions of Five Different Active Learning Strategies Used to Teach Software Modeling. *IEEE Access* 7: 184063-184077 (2019).
- Igor Scaliante Wiese, Rodrigo Takashi Kuroda, **Igor Steinmacher**, Gustavo Ansaldi Oliva, Reginaldo Ré, Christoph Treude, Marco Aurélio Gerosa. Pieces of contextual information suitable for predicting co-changes? An empirical study. *Softw. Qual. J.* 27(4): 1481-1503 (2019).
- **Igor Steinmacher**, Marco Aurélio Gerosa, Tayana Uchôa Conte, David F. Redmiles. Overcoming Social Barriers When Contributing to Open Source Software Projects. *Comput. Support. Cooperative Work.* 28(1-2): 247-290 (2019).
- **Igor Steinmacher**, Christoph Treude, Marco Aurélio Gerosa. Let Me In: Guidelines for the Successful Onboarding of Newcomers to Open Source Projects. *IEEE Softw.* 36(4): 41-49 (2019).
- Luis Felipe Dias, **Igor Steinmacher**, Gustavo Pinto. Who drives company-owned OSS projects: internal or external members? *J. Braz. Comput. Soc.* 24(1): 16:1-16:17 (2018).
- Mairieli Santos Wessel, Bruno Mendes de Souza, **Igor Steinmacher**, Igor Scaliante Wiese, Ivanilton Polato, Ana Paula Chaves, Marco Aurélio Gerosa. The Power of Bots: Characterizing and Understanding Bots in OSS Projects. *Proc. ACM Hum. Comput. Interact.* 2(CSCW): 182:1-182:19 (2018).
- Gustavo Pinto, **Igor Steinmacher**, Luiz Felipe Dias, Marco Aurélio Gerosa. On the challenges of open-sourcing proprietary software projects. *Empir. Softw. Eng.* 23(6): 3221-3247 (2018).
- Sogol Balali, **Igor Steinmacher**, Umayal Annamalai, Anita Sarma, Marco Aurélio Gerosa. Newcomers' Barriers. . . Is That All? An Analysis of Mentors' and Newcomers' Barriers in OSS Projects. *Comput. Support. Cooperative Work.* 27(3-6): 679-714 (2018).
- Anderson Bergamini de Neira, **Igor Steinmacher**, Igor Scaliante Wiese. Characterizing the hyperspecialists in the context of crowdsourcing software development. *J. Braz. Comput. Soc.* 24(1): 17:1-17:16 (2018).
- Igor Scaliante Wiese, Reginaldo Ré, **Igor Steinmacher**, Rodrigo Takashi Kuroda, Gustavo Ansaldi Oliva, Christoph Treude, Marco Aurélio Gerosa. Using contextual information to predict co-changes. *J. Syst. Softw.* 128: 220-235 (2017).

- **Igor Steinmacher**, Gregorio Robles, Brian Fitzgerald, Anthony I. Wasserman. Free and open source software development: the end of the teenage years. *J. Internet Serv. Appl.* 8(1): 17:1-17:4 (2017).
- Alexandre Lazaretti Zanatta, **Igor Steinmacher**, Letícia S. Machado, Cleidson R. B. de Souza, Rafael Prikladnicki. Barriers Faced by Newcomers to Software-Crowdsourcing Projects. *IEEE Softw.* 34(2): 37-43 (2017).
- **Igor Steinmacher**, Marco Aurélio Graciotto Silva, Marco Aurélio Gerosa, David F. Redmiles. A systematic literature review on the barriers faced by newcomers to open source software projects. *Inf. Softw. Technol.* 59: 67-85 (2015).
- Igor Scaliante Wiese, Douglas Nassif Junior, Reginaldo Ré, **Igor Steinmacher**, Marco Aurélio Gerosa. Comparing communication and development networks for predicting file change proneness: An exploratory study considering process and social metrics. *Electron. Commun. Eur. Assoc. Softw. Sci. Technol.* 65 (2014).
- **Igor Steinmacher**, Ana Paula Chaves, Marco Aurélio Gerosa. Awareness Support in Distributed Software Development: A Systematic Review and Mapping of the Literature. *Comput. Support. Cooperative Work.* 22(2-3): 113-158 (2013).
- Ana Paula Chaves, **Igor Steinmacher**, Gislaine Camila Lapasini Leal, Elisa Hatsue Moriya Huzita, Alberto B. Biasão. OntoDiSENv1: an Ontology to Support Global Software Development. *CLEI Electron. J.* 14(2) (2011).

Conference and Workshop Papers

- Nathan Cassee, Sankarsh Ravi, Ítalo Santos, **Igor Steinmacher**, Alexander Serebrenik. What barriers do students experience when trying to contribute to Open Source Software projects? *SEET@ICSE* , 151-161 (2026).
- Karl R. Wurst, Erik Falor, Avni Khatri, **Igor Steinmacher**. Ways to Make Git Easier to Teach. *SIGCSE* , 1627-1628 (2026).
- Mika Mäntylä, Patrícia Matsubara, Kátia Romero Felizardo, Miikka Kuutila, Marco Aurélio Gerosa, Savio Sousa Sampaio, Tayana Conte, **Igor Steinmacher**. Understanding LLMs in Title-Abstract Screening: From Disagreements to Recommendations. *SEAA* , 226-242 (2026).
- Sadia Afroz, Zixuan Feng, Tyler Menezes, Katie Kimura, Bianca Trinkenreich, **Igor Steinmacher**, Anita Sarma. The Fast and Spurious: Developer Productivity with GenAI. *SIGSOFT FSE Companion* , 1832-1841 (2026).
- Pawan Acharya, Tayana Conte, Jacob Penney, Pedro Oliveira, Misan Paul Etchie, Jared Duval, Bruno Gadelha, Marco Aurélio Gerosa, **Igor Steinmacher**. Shh! Don't Tell Them I Use GenAI: Exploring How Participants Use Generative AI in Hackathons. *SIGSOFT FSE Companion* , 1842-1850 (2026).
- Pedro Oliveira, Tayana Conte, Marco Aurélio Gerosa, **Igor Steinmacher**. Governance in Practice: How Open Source Projects Define and Document Roles. *CHASE@ICSE* , 255-266 (2026). ★ ACM SIGSOFT Distinguished Paper
- Raula Gaikovina Kula, Christoph Treude, Xing Hu, Sebastian Baltes, Earl T. Barr, Kelly Blincoe, Fabio Calefato, Junjie Chen, Marc Cheong, Youmei Fan, Daniel M. Germán, Marco Aurélio Gerosa, Jin L. C. Guo, Shinpei Hayashi, Robert Hirschfeld, Reid Holmes, Yintong Huo, Takashi Kobayashi, Michele Lanza, Zhongxin Liu, Olivier Nourry, Nicole Novielli, Denys Poshyvanyk, Shinobu Saito, Kazumasa Shimari, **Igor Steinmacher**, Mairieli Wessel, Markus Wagner, Annie Vella, Laurie Williams, Xin Xia. Forecasting Developer Environments with GenAI: A Research Perspective. *IDE@ICSE* , 1-5 (2026).
- Ashish Amresh, Siddharth Subramanian, **Igor Steinmacher**. Evaluating the Effect of Generative AI Content in Common Game Development Software Architectures. *GAS@ICSE* , 47-54 (2026).
- Rudrajit Choudhuri, Bianca Trinkenreich, Rahul Pandita, Eirini Kalliamvakou, **Igor Steinmacher**, Marco Aurélio Gerosa, Christopher Sanchez, Anita Sarma. What Guides Our Choices? Modeling Developers' Trust and Behavioral Intentions Towards Genai. *ICSE* , 1691-1703 (2025).

- Reine Santos, Gabriel da Silva Freitas, **Igor Steinmacher**, Tayana Conte, Ana Carolina Oran, Bruno Gadelha. User Stories: Does ChatGPT Do It Better? *ICEIS* , 47-58 (2025).
- Genildo Gomes, Paulo Silva, Sâmila Magalhães, Leandro Galvão, Tanara Lauschner, **Igor Steinmacher**, Tayana Conte, Bruno Gadelha. Understanding the Student Experience Better: Analyzing Remote Learning Tools Through Engagement, Participation, and Liveness Concepts. *CSEDU* , 958-965 (2025).
- Jacob Penney, Pawan Acharya, Peter Hilbert, Priyanka Parekh, Anita Sarma, **Igor Steinmacher**, Marco Aurélio Gerosa. Understanding Programming Students' Help-Seeking Preferences in the Era of Generative AI. *CompEd* , 15-21 (2025).
- Misan Paul Etchie, Hunter M. Beach, Kátia Romero Felizardo, **Igor Steinmacher**. Towards a Dataset to Assess Conversational Agent's Efficacy in Mentoring Newcomer Developers. *BotSE@ICSE* , 48-52 (2025).
- Zixuan Feng, **Igor Steinmacher**, Marco Aurélio Gerosa, Tyler Menezes, Alexander Serebrenik, Reed Milewicz, Anita Sarma. The Multifaceted Nature of Mentoring in OSS: Strategies, Qualities, and Ideal Outcomes. *CHASE@ICSE* , 203-214 (2025).
- Benjamin C. Carter, Jonathan Rivas Contreras, Carlos A. Llanes Villegas, Pawan Acharya, Jack Utzerath, Adonijah O. Farner, Hunter Jenkins, Dylan Johnson, Jacob Penney, **Igor Steinmacher**, Marco Aurélio Gerosa, Fabio Santos. SkillScope: A Tool to Predict Fine-Grained Skills Needed to Solve Issues on GitHub. *NLBSE@ICSE* , 9-12 (2025).
- Daniel Coutinho, Juliana Alves Pereira, Breno Braga Neves, João Lucas Correia, Caio Barbosa, Wesley K. G. Assunção, **Igor Steinmacher**, Marco Aurélio Gerosa, Augusto Baffa, Alessandro Garcia. PRemo: A Dataset of Emotions Found on Pull Request Discussions. *SBES* , 326-336 (2025).
- Jacob Penney, Pawan Acharya, Peter Hilbert, Priyanka Parekh, Anita Sarma, **Igor Steinmacher**, Marco Aurélio Gerosa. Outcomes, Perceptions, and Interaction Strategies of Novice Programmers Studying with ChatGPT. *CUI* , 73:1-73:15 (2025).
- Ítalo Santos, Kátia Romero Felizardo, Anita Sarma, **Igor Steinmacher**, Marco Aurélio Gerosa. Ossdoorway: A Gamified Environment to Scaffold Student Contributions to Open Source Software. *CSEE&T* , 325-335 (2025).
- Kátia Romero Felizardo, Anderson Deizepe, Daniel Coutinho, Genildo Gomes, Maria A. C. Meireles, Marco Aurélio Gerosa, **Igor Steinmacher**. On the Difficulties of Conducting and Replicating Systematic Literature Reviews Studies Using LLMs in Software Engineering. *WSESE@ICSE* , 20-23 (2025).
- Andressa S. S. Medeiros, Thelma Elita Colanzi, Igor Scaliante Wiese, Igor Fabio Steinmacher. O uso de exemplos trabalhados para facilitar a contribuição de novatos em projetos de software livre. *SBES* , 459-469 (2025).
- Bianca Trinkenreich, Zixuan Feng, Rudrajit Choudhuri, Marco Aurélio Gerosa, Anita Sarma, **Igor Steinmacher**. Investigating the Impact of Interpersonal Challenges on Feeling Welcome in OSS. *ICSE* , 717-729 (2025).
- Ana Maciel, Mairieli Wessel, Igor Wiese, **Igor Steinmacher**. Investigating the Impact of GitHub Discussions on Maintainers' Workload and Community Dynamics. *SBES* , 37-47 (2025).
- Rudrajit Choudhuri, Ambareesh Ramakrishnan, Amreeta Chatterjee, Bianca Trinkenreich, **Igor Steinmacher**, Marco Aurélio Gerosa, Anita Sarma. Insights from the Frontline: GenAI Utilization Among Software Engineering Students. *CSEE&T* , 1-12 (2025).
- Igor Scaliante Wiese, Jasmine Boyer, Ethan Rasgorshek, Gustavo Pinto, Marco Aurélio Gerosa, **Igor Steinmacher**, Bonita Sharif. How Developers Make Decisions When Choosing Issues and Reviewing Code: An Eye Tracking GitHub Study. *ETRA* , 2:1-2:7 (2025).
- Ítalo Santos, Kátia Romero Felizardo, **Igor Steinmacher**, Marco Aurélio Gerosa. Great Power Brings Great Responsibility: Personalizing Conversational AI for Diverse Problem-Solvers. *CHASE@ICSE* , 93-95 (2025).
- Pedro Oliveira, Doris Amoakohene, Toby Hocking, Marco Aurélio Gerosa, **Igor Steinmacher**. Governance Matters: Lessons From Restructuring the Data.Table OSS Project. *ICSME* , 1-12 (2025).

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Supervision

Ongoing Ph.D.

- Pedro A. Oliveira (2024–current), Northern Arizona University — open source governance as a foundation for project sustainability
- Jacob McAuley Penney (2023–current), Northern Arizona University — LLMs to support computational thinking and developer learning
- Morgan Nicholson (2025–current), Northern Arizona University — conversational agents in developer support

Concluded Ph.D.

- Ana Claudia Maciel (2026). Ph.D., Universidade Estadual de Maringá. *Investigating the Impact of GitHub Discussions on Maintainers' Workload and the Dynamics of Open Source Software Communities.* Supervisor.
- Italo Santos (2024). Ph.D., Northern Arizona University. *Supporting Diverse Problem-Solving Styles in Understanding How to Contribute to Open Source.* Co-supervisor (with Marco Gerosa).
- Simone de França Tonhão (2023). Ph.D., Universidade Estadual de Maringá. Supervisor.
- Fabio Santos (2023). Ph.D., Northern Arizona University. Co-supervisor (with Marco Gerosa).
- Marcia Lima (2023). Ph.D., Universidade Federal do Amazonas. Co-supervisor (with Tayana Conte and Bruno Gadelha).
- Bianca Trinkenreich (2022). Ph.D., Northern Arizona University. Supervisor.
- Patricia Matsubara (2022). Ph.D., Universidade Federal do Amazonas. Co-supervisor (with Tayana Conte).
- Mairieli Wessel (2021). Ph.D., Universidade de São Paulo. Co-supervisor (with Marco Gerosa).
- Williamson Silva (2020). Ph.D., Universidade Federal do Amazonas. Co-supervisor (with Tayana Conte).
- Jefferson Silva (2019). Ph.D., Universidade de São Paulo. Co-supervisor (with Marco Gerosa).
- Alexandre Lazaretti Zanatta (2018). Ph.D., Pontifícia Universidade Católica do Rio Grande do Sul. Co-supervisor (with Rafael Prikladnicki).

Concluded M.Sc.

- Misan Etchie (2024–2026). M.Sc., Northern Arizona University. *Improving Learning Outcomes Through a Gamified Software Engineering Learning Environment.* Supervisor.
- Pawan Acharya (2025). M.Sc., Northern Arizona University. *Exploring the Impact of LLM-Supported Collaboration on Team Dynamics and Participant Perceptions in Hackathons.* Supervisor.

- Jacob McAuley Penney (2024). M.Sc., Northern Arizona University. *Using Large Language Models to Support the Computational Thinking Learning Process.* Supervisor.
- Simone de França (2021). M.Sc., Universidade Estadual de Maringá. Supervisor.
- Carlos Danilo Luz (2021). M.Sc., Universidade Estadual de Maringá. Co-supervisor (with Edson Oliveira Jr.).
- Marcus Vinicius Bertoncello (2019). M.Sc., Universidade Estadual de Maringá. Supervisor.
- Anderson Bergamini de Neira (2018). M.Sc., Universidade Estadual de Maringá. Supervisor.
- Carolina Toscani (2018). M.Sc., Pontifícia Universidade Católica do Rio Grande do Sul. Co-supervisor (with Sabrina Marczak).
- João Choma Neto (2017). M.Sc., Universidade Estadual de Maringá. Co-supervisor (with Thelma Colanzi).

Keynotes, Invited Talks and Panels

- **2026** — Keynote, ICSE 2026 Doctoral Symposium (Rio de Janeiro, Brazil)
- **2026** — Panelist, ICSE 2026 Panels and Special Sessions: "SE.next: Raising the Agentic Engineer" (Rio de Janeiro, Brazil)
- **2026** — Invited talks, GiraEuropa 2026, university seminar tour (Italy & Switzerland)
- **2025** — Talk, FOSSY 2025, Science of Community track: "Lessons from a Decade of Open Source Sustainability Research" (Portland, OR, USA)
- **2025** — Panelist, BotSE 2025 (Workshop on Bots in Software Engineering) at ICSE 2025 (Ottawa, Canada)
- **2024** — Keynote, WBots 2024 (Workshop sobre Bots na Engenharia de Software) at CBSoft 2024 (Curitiba, Brazil)

Professional Service

Journal editorship

- Editorial Board — Empirical Software Engineering (since 2022)
- Editorial Board — Journal of Software: Evolution and Process (since 2021)
- Lead Guest Editor — SpringerOpen JISA, Thematic Series on Free and Open Source Software Development
- Co-Guest Editor — IEEE Software Thematic Series: 20 Years of Open Source

Conference organization

- Student Mentoring Workshop @ ICSE (SMeW 2027)
- General Chair — International Conference on Mining Software Repositories (MSR 2026)
- General Chair — IEEE International Conference on Software Maintenance and Evolution (ICSME 2024)
- General Co-Chair — International Conference on Program Comprehension (ICPC 2024)
- General Chair — International Conference on Cooperative and Human Aspects of Software Engineering (CHASE 2023)
- Virtualization Co-Chair — ICSSP-ICGSE 2021
- Co-organizer — Workshop on Cooperative and Human Aspects of Software Engineering (CHASE 2019)
- Organizer — AZ Computing Postdocs Best Practices (2017)
- General Co-Chair — Brazilian Conference on Software: Theory and Practice (CBSoft 2016)
- Co-organizer — Workshop sobre Aspectos Sociais, Humanos e Econômicos de Software (WASHES 2016)
- General Chair — Free/Libre Open Source Software Workshop (WSL 2014, 2015)
- Organizing Committee — Brazilian Symposium on Collaborative Systems (SBSC 2012)

Program committee chair

- Area Chair, Human Aspects — ICSE 2025
- Program Co-Chair — Brazilian Symposium on Software Engineering, Main Track (SBES 2024)
- Program Co-Chair — Brazilian Symposium on Software Engineering, Insightful Ideas (SBES IIER 2023)
- Program Co-Chair — 18th IFIP Conference on Open Source Systems (OSS 2022)
- Program Co-Chair — International Conference on Software and Systems Process (ICSSP 2022)
- Program Co-Chair — Brazilian Symposium on Software Engineering, Tools Track (SBES Tools 2022)
- Co-Chair — ICSE Student Contest on Software Engineering (SCORE 2021)
- Program Co-Chair — ACM/IEEE International Conference on Global Software Engineering (ICGSE 2020)
- Program Co-Chair — ACS/IEEE International Conference on Computer Systems and Applications (AICCSA 2019)

Program committee membership

- ICSE: International Conference on Software Engineering (2022, 2023, 2024, 2025)
- MSR: International Conference on Mining Software Repositories (2023, 2024)
- EASE: International Conference on Evaluation and Assessment in Software Engineering (2017, 2018, 2024)
- GROUP: ACM International Conference on Supporting Group Work (2020)
- ICSE-SEIS: Software Engineering in Society track (2020)
- SANER Tool Track: International Conference on Software Analysis, Evolution and Reengineering (2019, 2020)
- BotSE: International Workshop on Bots for Software Engineering (2019, 2020, 2022)
- CSCW: ACM Conference on Computer Supported Cooperative Work and Social Computing (2019, 2020)
- ICSE-NIER: New Ideas and Emerging Results track (2019)
- ICGSE: IEEE International Conference on Global Software Engineering (2016, 2017, 2018, 2019)
- SBES: Brazilian Symposium on Software Engineering (2017, 2018, 2019)
- OSS: International Conference on Open Source Systems (2017, 2018)
- ICSME Artifacts Track (2017)
- CHASE: International Workshop on Cooperative and Human Aspects of Software Engineering (2017, 2018)
- CIbSE: Iberoamerican Congress of Software Engineering (2016, 2017)
- SBSC: Brazilian Symposium on Collaborative Systems (2015, 2016, 2017)
- VEM: Workshop on Software Visualization, Evolution, and Maintenance (2015, 2016, 2017)

Journal reviewing

- IEEE Transactions on Software Engineering; ACM TOSEM; ACM Computing Surveys; Communications of the ACM; Empirical Software Engineering; Information and Software Technology; Journal of Systems and Software; Journal of Software: Evolution and Process; Computer Supported Cooperative Work; Computer Standards & Interfaces; Journal of the Brazilian Computer Society; IET Software
- Distinguished Reviewer Award — ICSE 2023

Other service

- Associate Director for Graduate Programs and Academic Affairs — SICCS, NAU
- Chair, proposal committee for NAU's BS in Artificial Intelligence (starts Fall 2026)
- Proceedings Chair — ICSME 2022
- Publicity Chair — ICSME 2018
- Publicity Chair for South America — OSS 2018
- Publicity and Social Media Chair — ICGSE 2017

- Proceedings Chair — ICGSE 2016
- Member of the Education Committee — Brazilian Computer Society (2015–2017)
- Member of the Collaborative Systems Special Interest Group (CESC) — Brazilian Computer Society (2015–2016)

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